

MILESTONE 1 PRESENTATION

AI Smart Bug Analyzer & Fix Advisor

Core operational ingestion, RAG similarity workflows, and data schema blueprints for automated defect lifecycle triaging.

PLATFORM DEVELOPMENT DECK

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AUDIENCE: DEVELOPERS & STAKEHOLDERS

Project Overview

Operational Pain Point

Manual bug triaging, root-cause analysis, and duplicate detection are highly time-consuming and error-prone processes.

Developers lose standard sprint cycles parsing disorganized stack traces, running duplicate queries, or rebuilding hotfixes that have already been resolved elsewhere in old catalog folders.

Milestone 1 Objectives

We establish the absolute operational foundation by building the core ingestion channel and seeding the historical dataset index.

- ✓ Functional Bug Ingestion Interface.
- ✓ ChromaDB Persistent Storage.
- ✓ 5,000 public dataset reports cleaned.

Detailed Delivery Requirements

1. Research & Blueprinting

Analyze defect triage schemas, semantic distance functions, RAG pipeline window segment limits, and multi-repository issue tracker syntax rules.

2. Core Implementation

Establish a solid functional codebase rather than purely academic notes. Write functional code to clean, index, and query historical trace logs.

Milestone 1 Deliverables Checklist

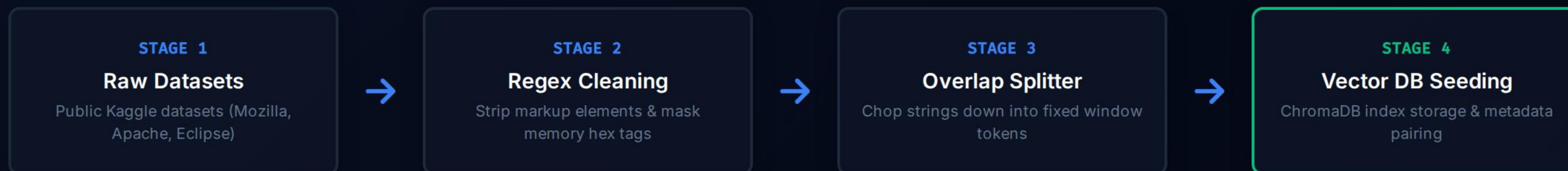
- ✔ Complete Architecture Designs
- ✔ Documented Multi-Agent Specs
- ✔ Streamlit Bug Submission UI
- ✔ Preprocessed & Chunked Datasets
- ✔ Populated ChromaDB Vector Index

Technology Stack

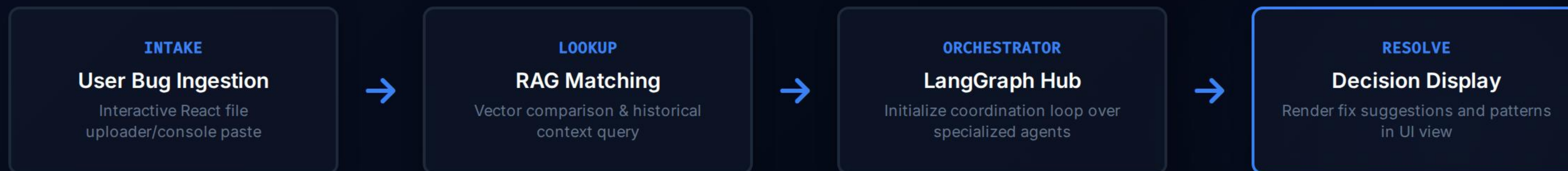
LAYER	SELECTED TECH	PURPOSE & FUNCTIONAL JUSTIFICATION
Frontend UI	React.js / Tailwind CSS	Renders interactive dashboards, submission panels, file handling states, and structured diagnostic displays.
Backend REST API	FastAPI (Python)	Asynchronous routes, ultra-low payload parsing latencies, and simplified bindings for ML architectures.
Agent Orchestration	LangGraph (LangChain)	Stateful multi-agent looping structures, conditional routing, and context orchestration.
Vector Store	ChromaDB (Local)	High-speed similarity calculations on cosine spaces, running completely in persistent disk memory.

System Architecture

I. OFFLINE DATA KNOWLEDGE INGESTION ROADWAY



II. RUNTIME ONLINE ANALYTICAL HANDSHAKE



Agent Responsibilities



Triage Agent

Parses bugs, evaluates severity, isolates target components.



Log Analyzer

Extracts crash classes, maps exception bounds inside code lines.



Duplicate Agent

Queries vector similarity maps, checks active duplication issues.



Root Cause Agent

Analyzes contextual data variables, outputs theoretical breakdown.



Remediation Agent

Synthesizes standard software fix designs, updates libraries.

Knowledge Base Schema Design

bug_id

string (PK)

source_repository

string (indexed)

component

string

severity

string

raw_description

string (full-text)

Vector-to-Metadata Index Pairing

By integrating explicit categorical parameters alongside multidimensional document arrays, our pipeline triggers highly selective vector filtering.

This eliminates semantic bleed, allowing searches to narrow scope immediately to repository segments (e.g. only matching *Mozilla* vector trees).

Bug Submission Module

Flexible Log Submission Portal

Built using Streamlit/React, our active user application provides developers with a dedicated, zero-friction error submission form.

Supports instant copy-paste strings, console stack trace formats, and large continuous system files (.txt / .log extensions).

http://localhost:8501/bug_submission

AI Smart Bug Analyzer

Bug Submission Portal • Milestone 1 Ingest

Paste Bug Report

Title: Login API throws NullPointerException

Environment: Java 17 Production Build

Upload Stack Trace or Error Log

 error.log.txt (0.6 KB) Uploaded

Submit Log Trace

Data Preprocessing & Cleaning

Optimizing Raw Log Inputs

Unprocessed stack traces contain high-volume semantic noise, memory pointer shifts, system formatting wrappers, and raw trace markers.

To prevent vector skew, our pipeline applies strict cleansing regex strategies:

-  Strips markup tags and structural JSON envelopes.
-  Replaces volatile raw hex pointers with generic tags.
-  Eliminates formatting whitespace.

python_pipeline_preprocessor.py

Active

```
user@dev:~$ python preprocess_datasets.py
[INFO] Locating Kaggle raw inputs...
[INFO] Detected: Mozilla, Apache, Eclipse datasets.
[PROCESS] Running regex HTML stripper...
[PROCESS] Masking raw hex stack pointer strings...
[SUCCESS] Successfully processed 5,000 files.
[SUCCESS] Standardized space arrays mapped to database memory.
```


Recursive Text Chunking

recursive_splitter.py

```
>>> from langchain.text_splitter import RecursiveCharacterTextSplitter
>>> splitter = RecursiveCharacterTextSplitter(chunk_size=750,
chunk_overlap=75)
>>> chunks = splitter.split_text(cleaned_report)
>>> print(f"Total Chunks Created: 5000")
```

Why 10% Overlap Bounds?

Ensures trace lines, context indicators, and function parameter stacks do not get separated over segment divisions.

Splitting Boundary Strategy

Continuous error logs and descriptions vary in shape and length, challenging context parameters during embedding processing.

We apply a **750-character window** with a **75-character sliding overlap**. This matches context lengths and extracts dense, highly structured blocks.

Our pipeline generated exactly **5,000 document segments** from public Kaggle inputs, establishing a robust, localized search index.

Embedding Generation

Selected Model: **all-MiniLM-L6-v2**

Vector Dimension: **384 Dimensions**

Indexed Datasets: **5,000 Reports**

Embedding Pipeline Execution

Our pipeline processes the cleaned text segments into dense vector coordinates using HuggingFace's local **all-MiniLM-L6-v2** model.

This establishes a mathematical similarity space, mapping related crash anomalies to neighboring vector planes.

```
bash (Terminal 3) - AI-Smart-Bug-Analyzer/rag
```

```
Github: https://github.com/saibhanahammed/AI-Smart-Bug-Analyzer
```

```
(venv) user@HomePC:~/AI-Smart-Bug-Analyzer/rag$ python preprocess.py
```

```
Preprocessing Complete
```

```
(venv) user@HomePC:~/AI-Smart-Bug-Analyzer/rag$ python chunk.py
```

```
Chunking Complete. Total Chunks Created: 5000
```

```
(venv) user@HomePC:~/AI-Smart-Bug-Analyzer/rag$ python embed.py
```

```
Loading model ...
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits.
```

```
[SUCCESS] Successfully generated 5000 vector objects. Committing to ChromaDB persistent collection...
```


Vector Database Indexing

Vector Persistence & Storage

ChromaDB serves as our local vector database, running directly in persistent disk storage.

We configure our search indexes with **HNSW Cosine Distance** models, optimizing the balance between speed and similarity precision.

During lookup cycles, the backend retrieves similar cases in under 50ms, delivering immediate historical context back to our agents.

database_config.py

```
>>> import chromadb
>>> client = chromadb.PersistentClient(path="./chroma_db")
>>> collection = client.create_collection(
            name="historical_defects",
            metadata={"hnsw:space": "cosine"}
        )
[DATABASE] persistent folder loaded correctly. Index ready.
```


Thank You

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 [saibhanahammed/AI-Smart-Bug-Analyzer](https://github.com/saibhanahammed/AI-Smart-Bug-Analyzer)

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